

Saketh Ayyagari

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EDUCATION

University of Maryland, College Park

College Park, MD

Bachelor of Science: Computer Science, Minor in Robotics

May 2028

EXPERIENCE

Machine Learning Engineer

Feb 2026 - May 2026

Children's National Hospital

College Park, MD

- Developed a mobile application to detect safe vs. unsafe football tackles using pose estimation algorithms to support patient safety and research operations.
- Designed recording and clip-analysis features in Figma in collaboration with 5 full-stack engineers.
- Annotated and analyzed 70+ videos of football plays using Roboflow to train and evaluate a pose estimation model.

Computer Vision and Navigation Software Developer

Sep. 2025 - Present

Robotics at Maryland

College Park, MD

- Collaborating with mechanical, electrical, and software subteams to develop a Mars Rover for Canadian International Rover Challenge.
- Integrated depth camera into a Gazebo simulation to increase perception capabilities by obtaining PointCloud stream.
- Developed ROS2 packages utilizing point cloud data to generate artificial potential field to avoid obstacles when planning paths.

File Systems Intern

Sep. 2024 - Jan. 2025

Commvault

Tinton Falls, NJ

- Developed Python script to visualize file metadata changes across 100+ files over a 7-day interval.
- Implemented anomaly detection functionalities using existing C++ code, file metadata, and SQL queries.
- Tested false positive detection methods to improve script reliability and accuracy.

Research Intern

Jul. 2024 - Aug. 2024

MIT Beaver Works Summer Institute

Cambridge, MA

- Selected as 1 of 30 students for the Autonomous RACECAR Project.
- Developed computer vision and control algorithms with ROS2 to navigate a scale-model car; placed 3rd out of 7 teams in Grand Prix.
- Published research paper on IMU-based tire blowout detection; accepted to 2025 IEEE Vehicular Technological Conference.

PROJECTS

Simultaneous Localization and Mapping System

Feb. 2025 - Jun. 2025

- Fused LiDAR and odometry data from a robot for accurate localization and map generation of unknown environments using slam-toolbox.
- Integrated encoders, IMU, 2D LiDAR and Raspberry Pi 4B into a physical differential drive robot.
- Designed motor/sensor interfaces and teleoperation pipelines in ROS2 Python for real-time robot control and data logging.

Predicting Movement Direction with Machine Learning

Mar 2024 - Jan 2025

- Trained SVMs and Random Forests using scikit-learn, and Neural Networks with Tensorflow on neural spike data to predict the direction of a mammal's movement.
- Tested accuracy and loss of four models on a dataset of 143 neurons with spike data over 158 trials.
- Drafted a 7-page research paper outlining methodology and results.

Neural Network for Pneumonia Detection

Jul 2023

- Built convolutional neural networks with Tensorflow for detecting pneumonia with 83% peak accuracy.
- Augmented dataset of 100+ images of chest X-Rays with rotations, scalings, and color changes with OpenCV to improve accuracy from 70% to 83%.
- Applied transfer learning techniques by implementing VGG16 pretrained model to improve accuracy
- Deployed model to Python-based web application with Streamlit for users to test.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, C

Frameworks/Libraries: Tensorflow, PyTorch, NumPy, Scikit-learn, ROS2, Gazebo, OpenCV, Streamlit.

Tools: Git, GitHub, Docker, VS Code, Visual Studio, PyCharm, Eclipse, Roboflow, Microsoft Office, Google Suite